

FraudGuard: Fake Job Listing Detection using Deep Learning and Agentic AI

Data Science and AI Lab Project – Final Presentation

Group: Group 9

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The Problem: Fake Job Listings at Scale



3-5% of all online job postings are fraudulent.

DATA METRIC



Modern fraud postings mimic legitimate ads with linguistic sophistication — keyword filters fail.



Victims lose money, personal data, and time — real-world impact on Indian job seekers.

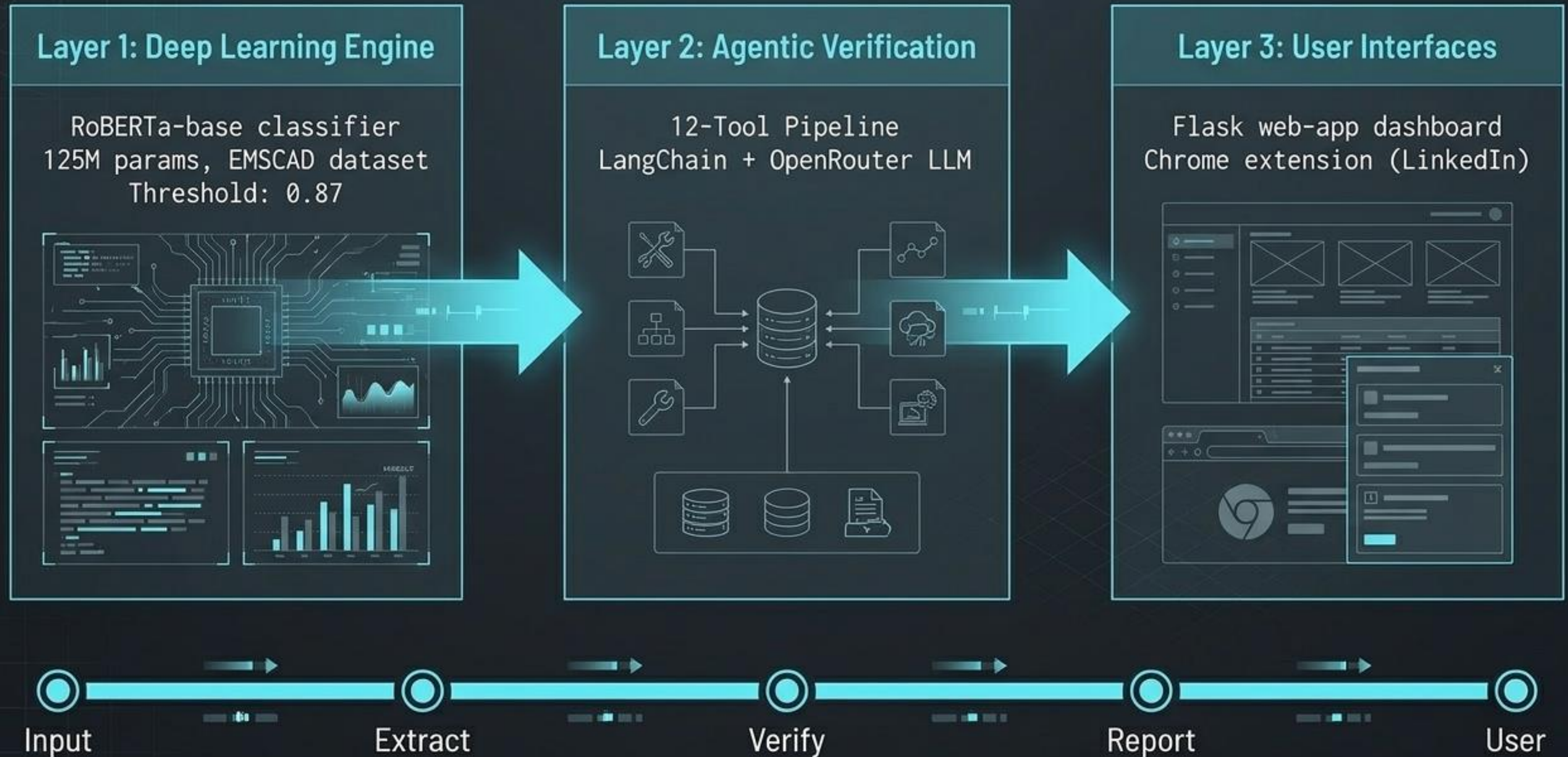


No existing tool combines deep NLP + external verification + human-readable explanation.



FraudGuard doesn't just output a probability score;
it **cross-references** the web to prove its verdict.

FraudGuard – Three-Layer Architecture



The Objective and the 20:1 Antagonist

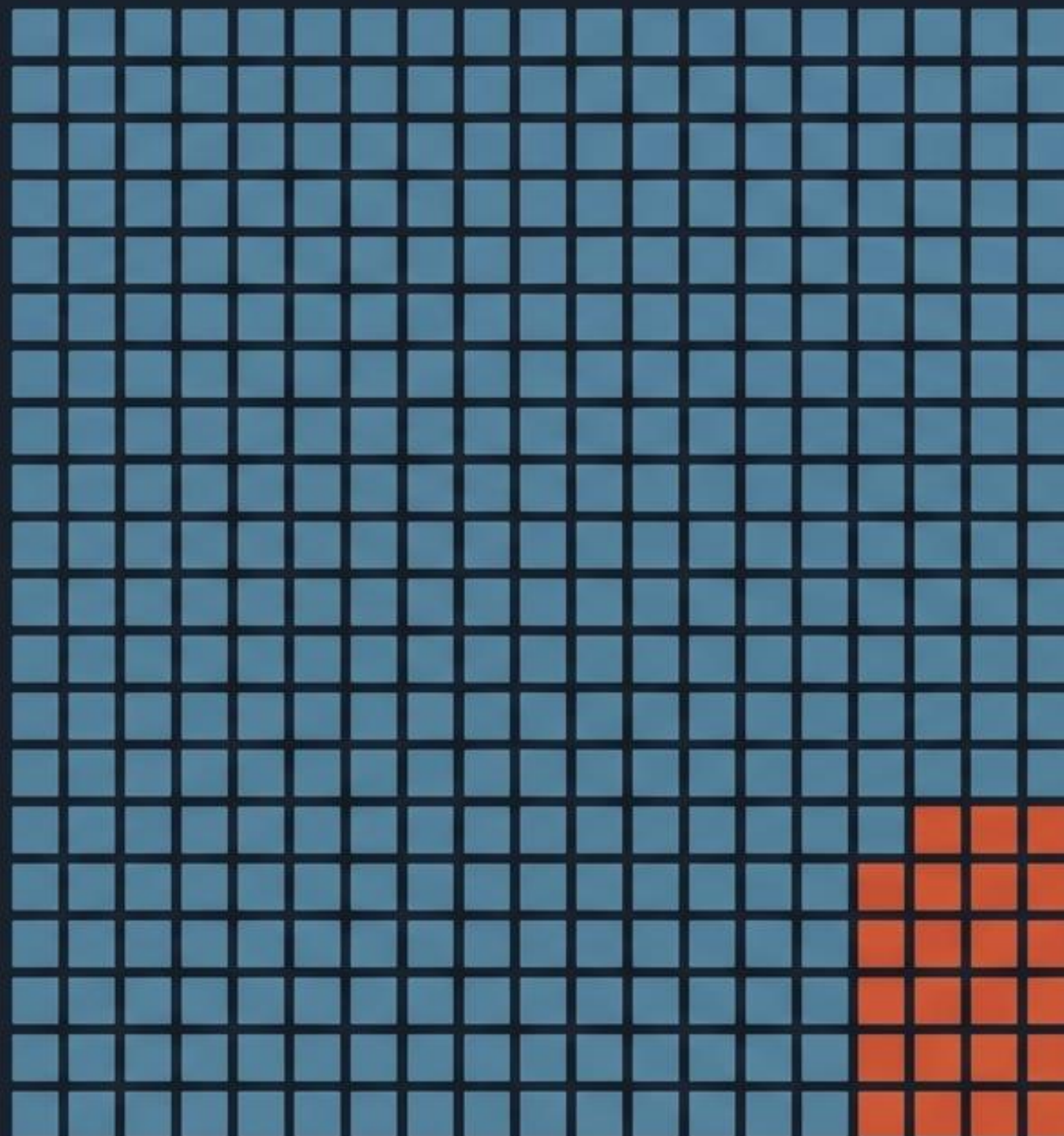
Goal: Build an AI model to automatically detect fraudulent job postings.

Unstructured Inputs:

Title, Description, Requirements,
Company Profile, Benefits.

Structured Inputs:

Location, Department, Salary
Range, Employment Type, Industry,
Logo Presence.



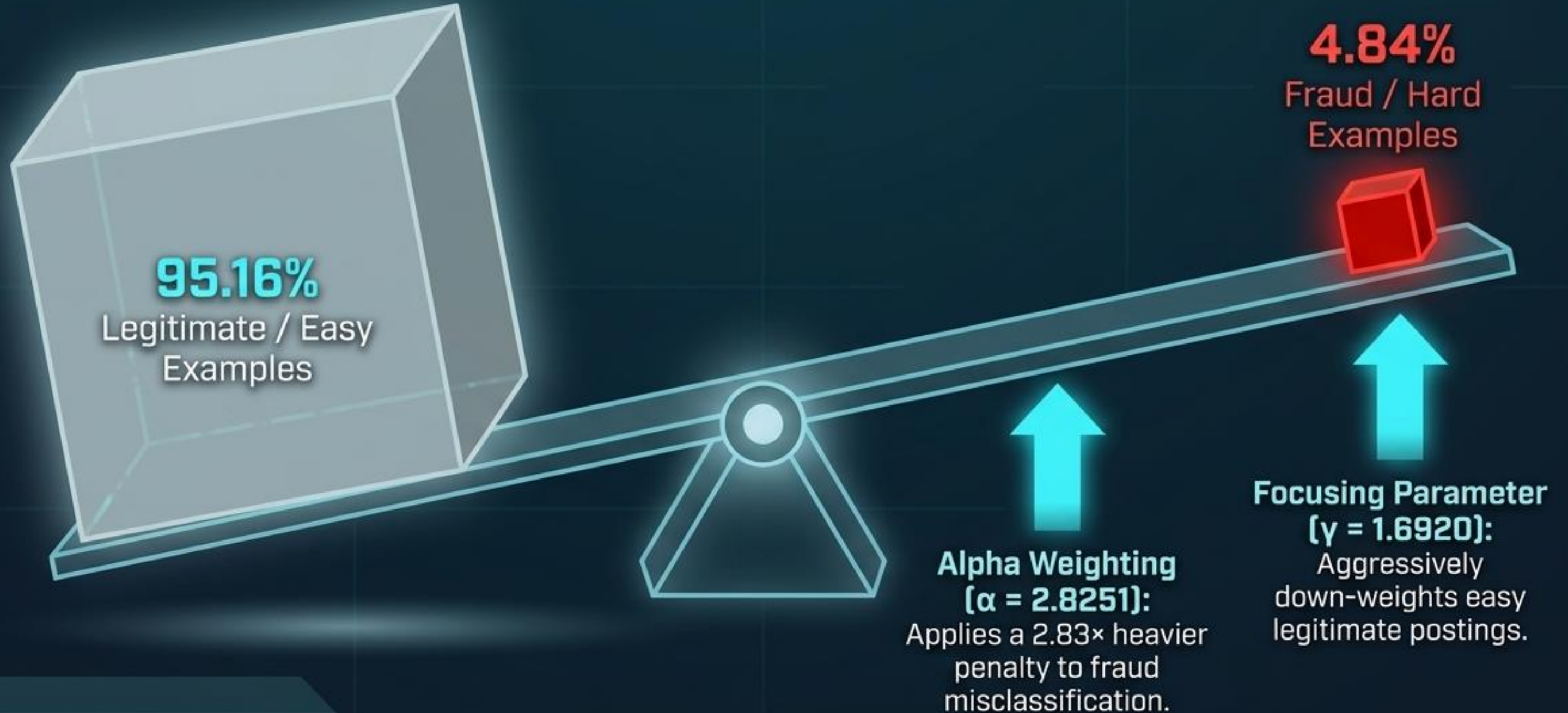
17,880

Total
Listings

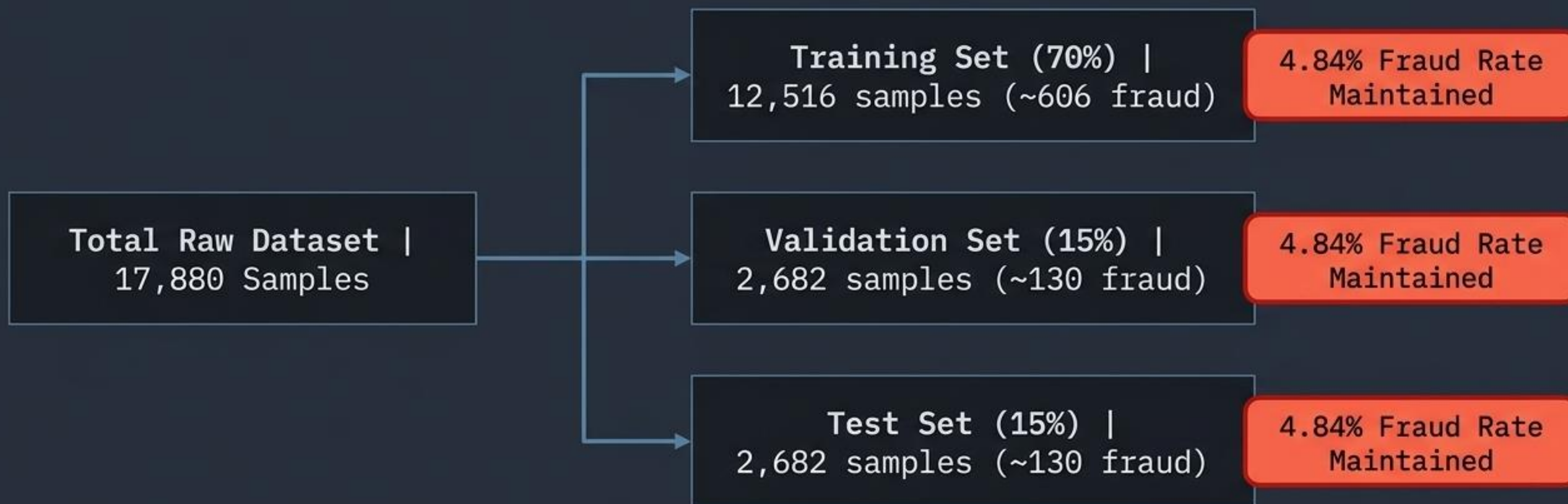
95.2%
Legitimate

4.8%
Fraudulent

OVERCOMING IMBALANCE: FOCAL LOSS MECHANISM

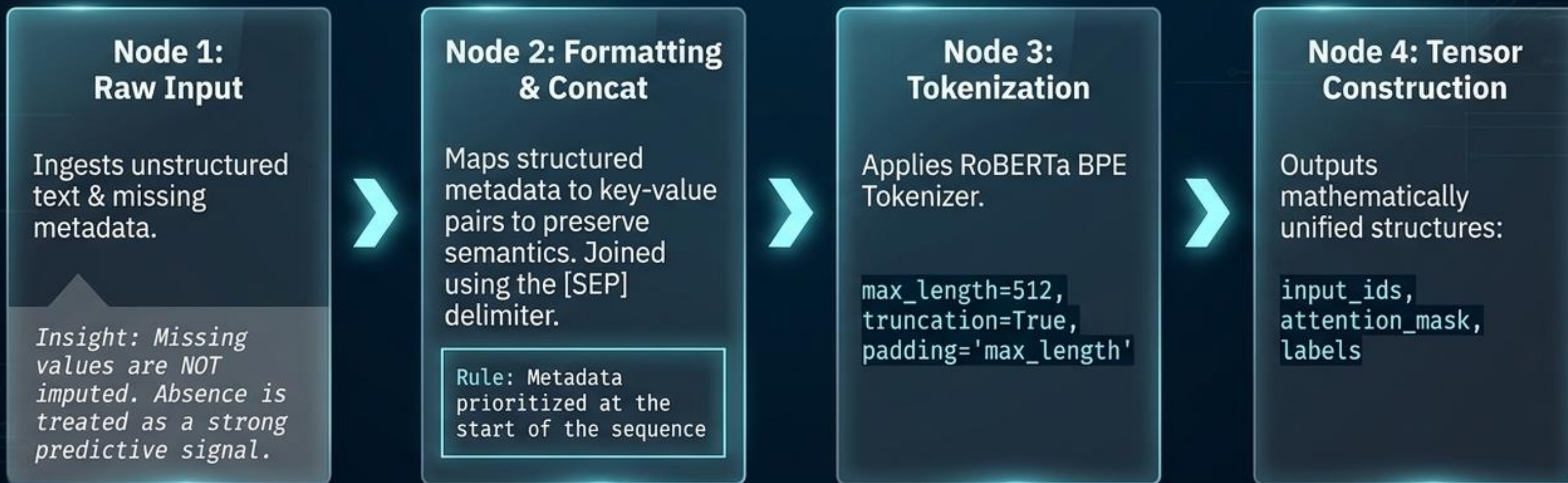


Dataset Organization & Rigorous Stratification



Insight: Stratified sampling guarantees the extreme 20:1 class imbalance is perfectly preserved across all splits, ensuring evaluation metrics reflect true, real-world deployment conditions.

Preprocessing Flow: Metamorphosis to Tensors



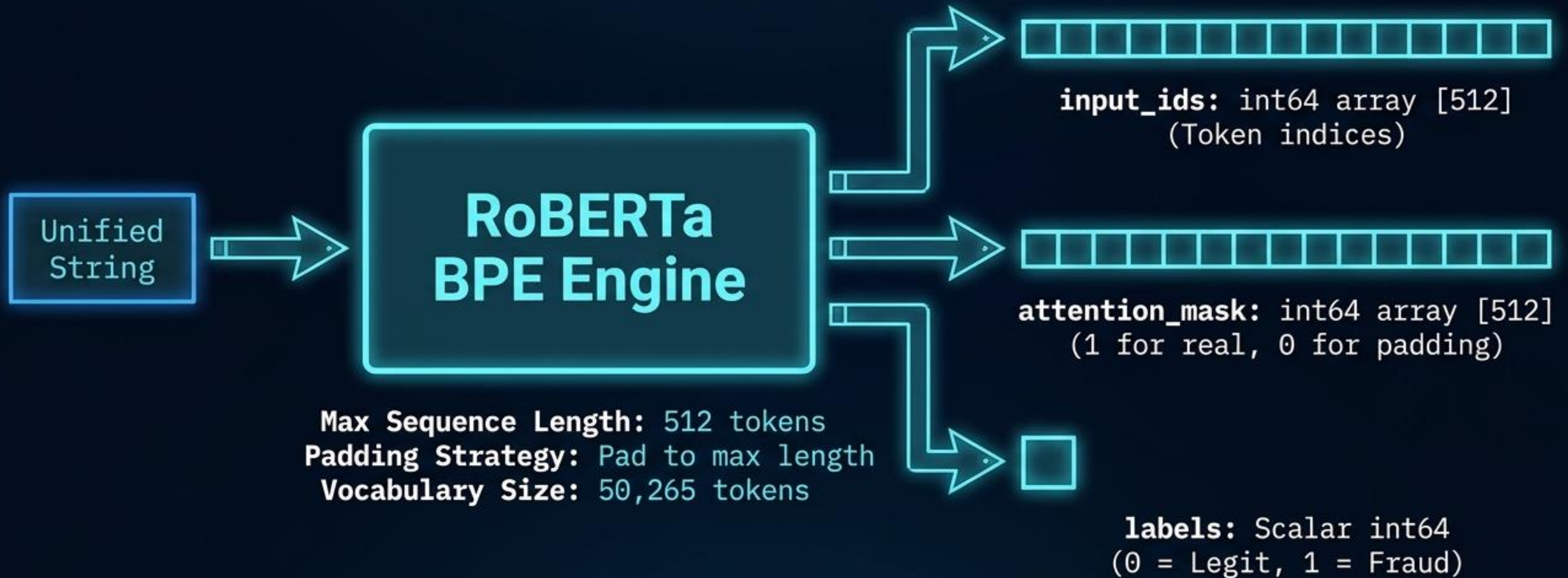
This pipeline guarantees heterogeneous signals (salary, title, raw description) are converted into a unified contextual sequence expected by the transformer.

Preprocessing Innovation: The [SEP] Strategy



Structured metadata fields are placed FIRST in the [SEP]-joined input string.
This protects critical signals from 512-token truncation.

TOKENIZATION & TENSOR CONSTRUCTION



Module: Input Sequence Construction

```
STRUCT_COLS = ['location', 'department', 'salary_range',  
'employment_type', 'required_experience', 'required_education',  
'industry', 'function', 'has_company_logo']  
TEXT_COLS = ['title', 'description', 'requirements',  
'company_profile', 'benefits']
```

```
def build_input_text(row):  
    parts = []  
    for col in STRUCT_COLS:  
        val = str(row.get(col, '') or '').strip()  
        if val and val.lower() not in ('nan', 'none', ''):  
            parts.append(f"{col.replace('_', ' ').title()}:  
{val}")  
    for col in TEXT_COLS:  
        val = str(row.get(col, '') or '').strip()  
        if val and val.lower() not in ('nan', 'none', ''):  
            parts.append(val)
```

```
[return ' [SEP] '.join(parts)]
```

```
df['input_text'] = df.apply(build_input_text, axis=1)  
df['label'] = df['fraudulent'].astype(int)
```

Module: RoBERTa Tokenization

```
# Module: RoBERTa Tokenization
```

```
from transformers import AutoTokenizer
```

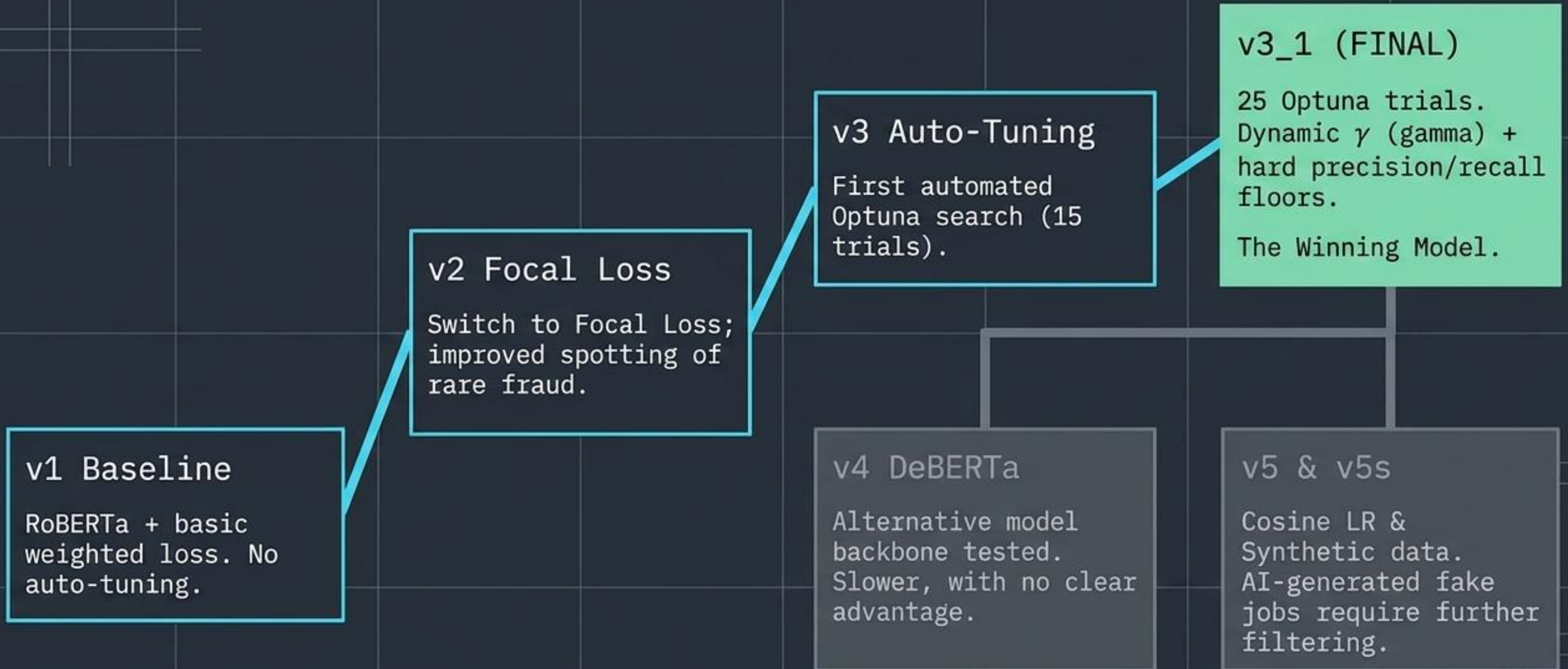
```
MODEL_NAME = 'roberta-base'
```

```
MAX_SEQ_LEN = 512
```

```
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)
```

```
def tokenize_fn(batch):  
    return tokenizer(batch['input_text'],  
                      truncation=True,  
                      padding='max_length',  
                      max_length=MAX_SEQ_LEN)
```


System Evolution Map



The Strategic Arsenal



Focal Loss & Class Weighting

Mathematically forces the model to focus on difficult, rare fraud examples.

Penalises the model **20x more** for missing a fraudulent post than for a false alarm.



Optuna HPO

Replaces manual guesswork with an automated **25-trial search algorithm**.

Hunts for the optimal combination of learning rates and dynamic gamma values.



Threshold Tuning

Abandons the default 50% probability cut-off.

Dynamically locates the perfect mathematical boundary to balance precision against recall.

25-Trial Optuna HPO — Systematic Search with Hard Floors

lr: 2.59e-5	warmup: 0.1506	batch: 16	weight_decay: 0.0702
focal_gamma: 1.6920	fraud_class_weight: 2.8251		epochs: 9 (stopped at 7)

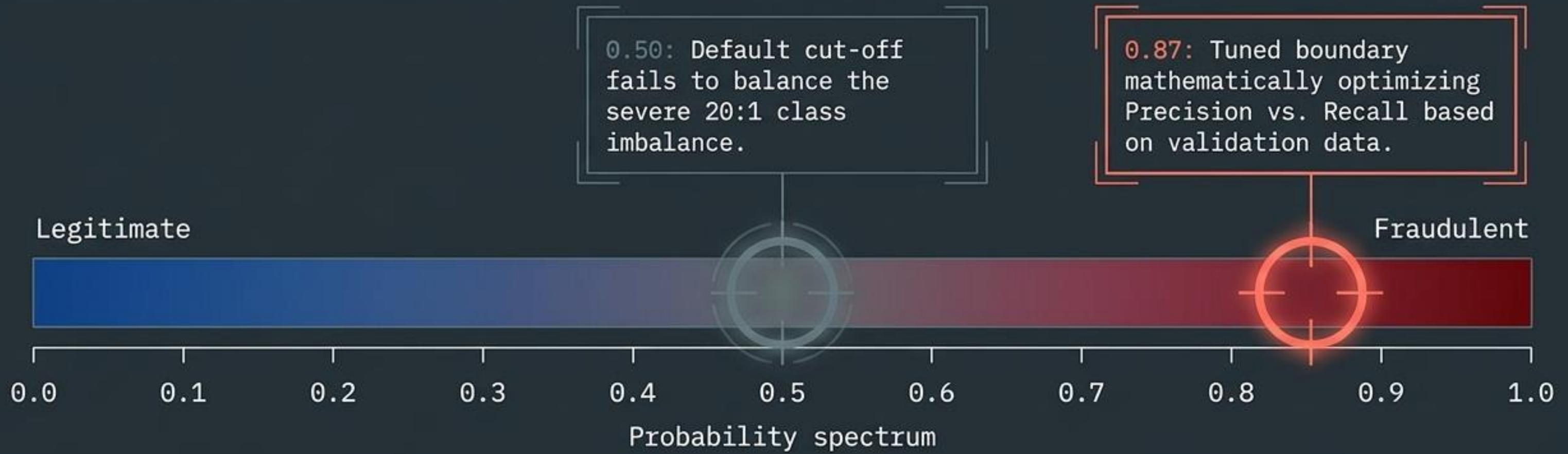
Key Innovation:

Trials are PRUNED unless
Recall ≥ 0.89 **AND** Precision ≥ 0.93
simultaneously.

This prevents optimizing F1 at the
expense of either metric.



Visualizing the Boundary: Threshold Tuning



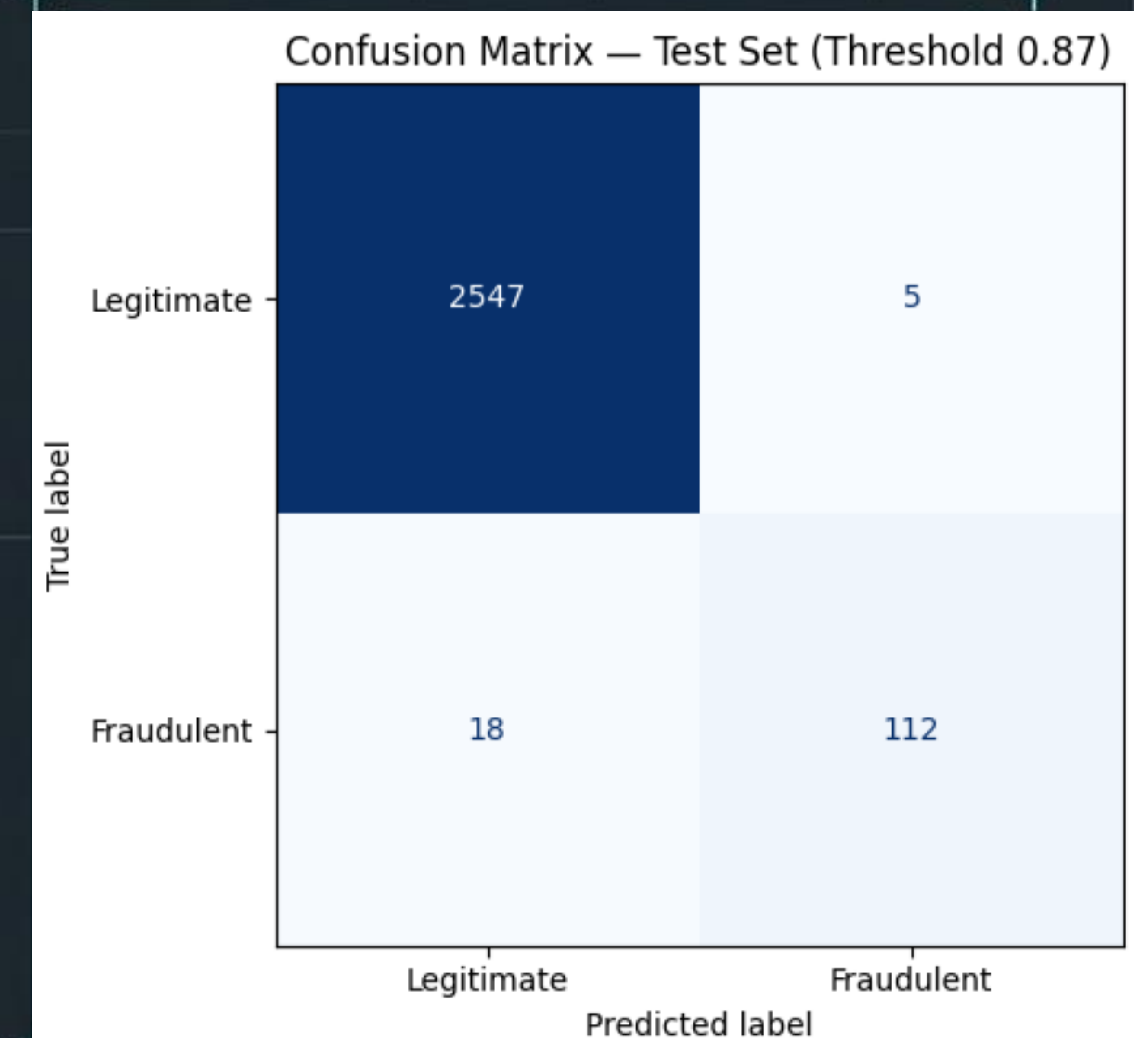
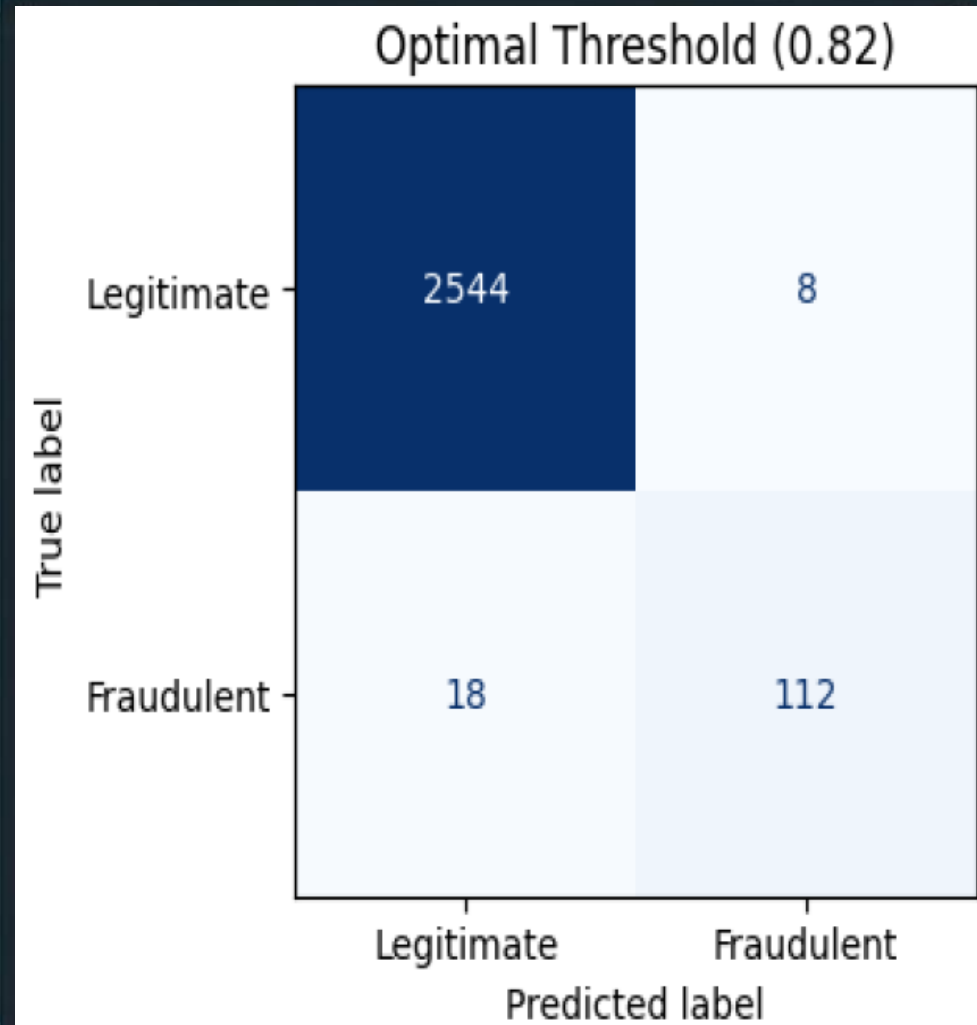
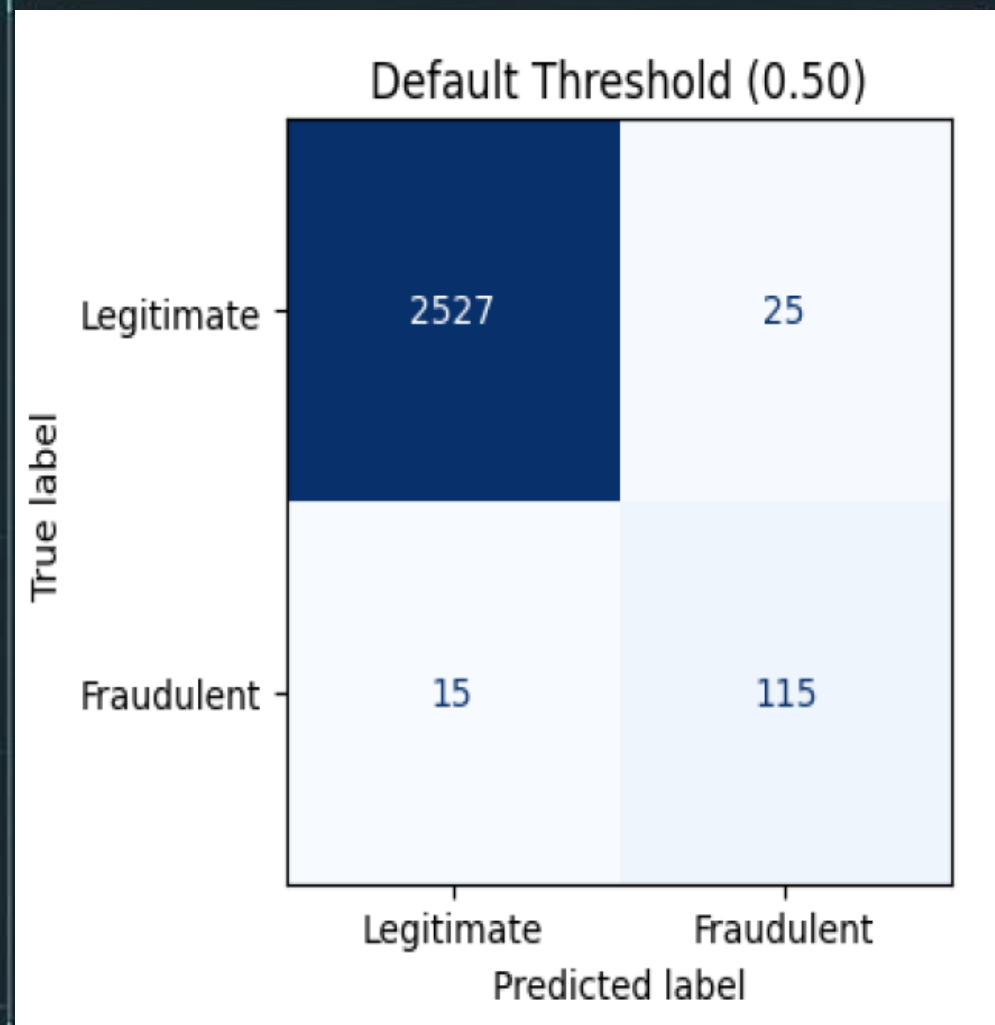
Key Insight

The RoBERTa model does not output a simple Yes/No. It outputs a continuous probability spectrum. Moving the decision boundary to 0.87 forces the model to be highly confident before flagging a job, protecting precision.

TERM DICTIONARY

Probability Threshold: The exact numerical cutoff point where the model's prediction flips from Legitimate to Fraudulent.

Forensic Diagnostic Overlays



Diagnostic Insight

By raising the threshold to 0.87, we successfully crushed false alarms (dropping from 25 to 5), ensuring high platform trust, at the calculated expense of letting 3 additional fraudulent posts slip through.

F1-Score (Primary Target)

The ultimate balancer.
Severely penalizes models
that sacrifice one metric
for the other.

Recall (Safety Target)

Fraction of actual fraud
caught. Missing fraud creates
real-world victims.

Warning:
Standard Accuracy
is useless here.
Predicting 'all
real' blindly
yields 95.1%
accuracy.

Precision (Trust Target)

Fraction of predicted fraud
that is actually fraud.
Flagging legitimate jobs
destroys platform trust.

ROC-AUC (Separation Target)

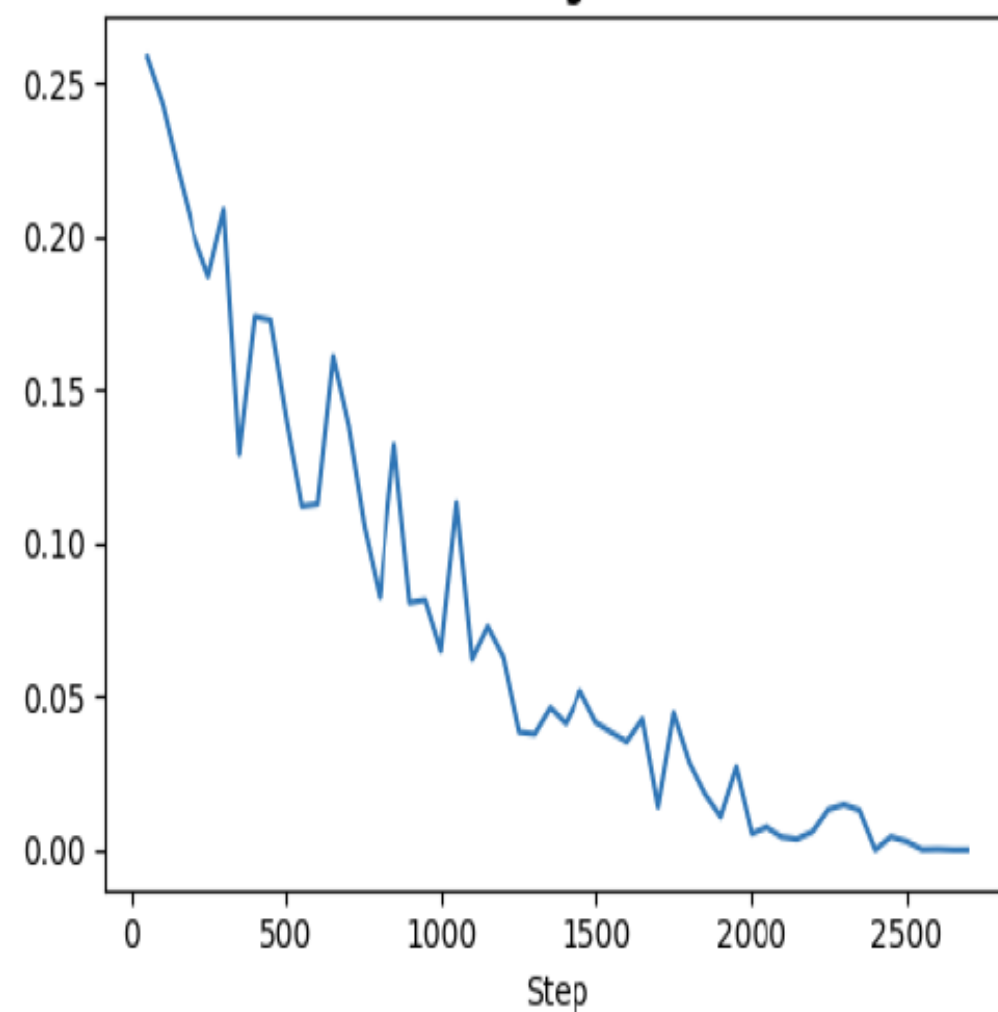
Overall ranking quality
regardless of the chosen
threshold.

Quantitative Results: The Scorecard

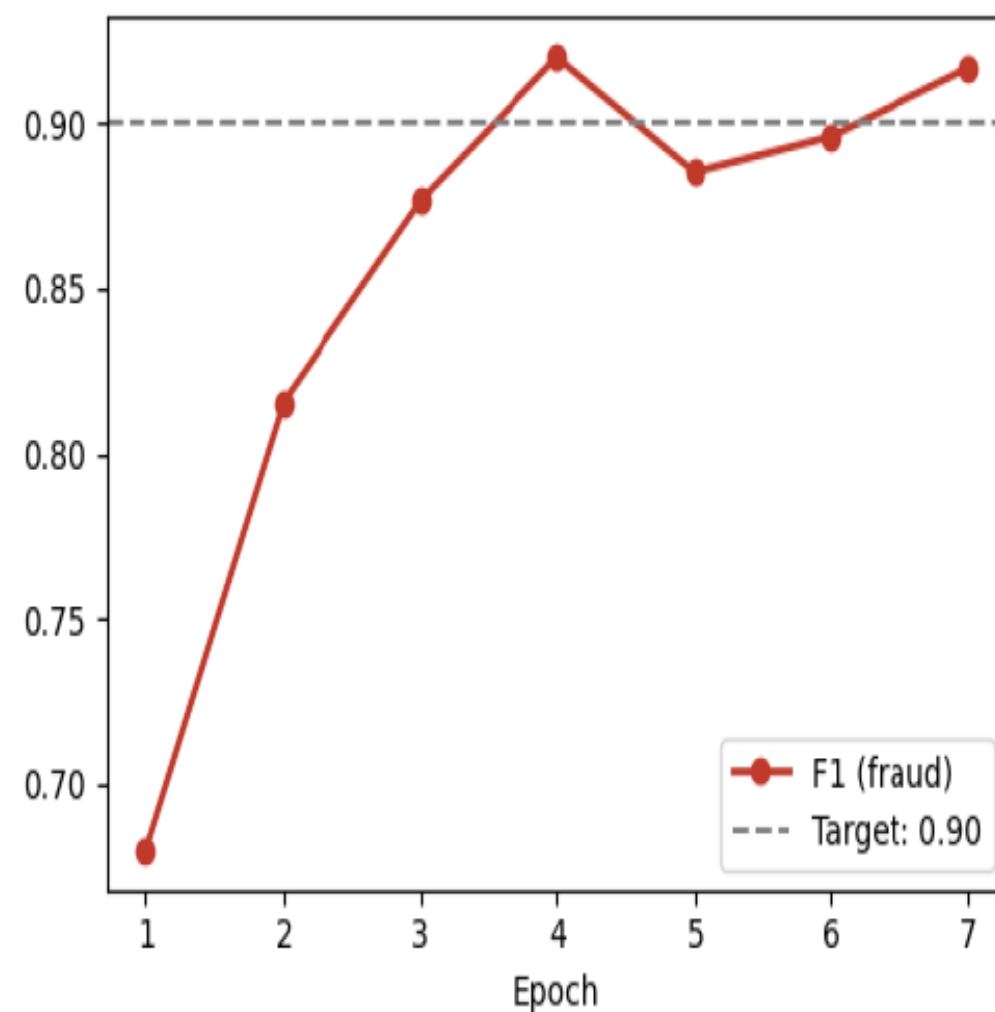
Metric (Target)	v1 Baseline	v3_1 Final	Gap-delta
Precision (Target: ≥ 0.93)	0.920	0.9573	↑ [Target Exceeded: +0.027]
ROC-AUC (Target: ≥ 0.95)	0.987	0.9930	↑ [Target Exceeded: +0.043]
F1-Score (Target: ≥ 0.91)	0.874	0.9069	← [Narrow Miss: -0.003]
Recall (Target: ≥ 0.89)	0.830	0.8615	← [Miss: -0.028]

Insight: The model dynamically maximizes precision and overall accuracy, conquering the severe imbalance to come within fractions of a percent of the primary F1 target.

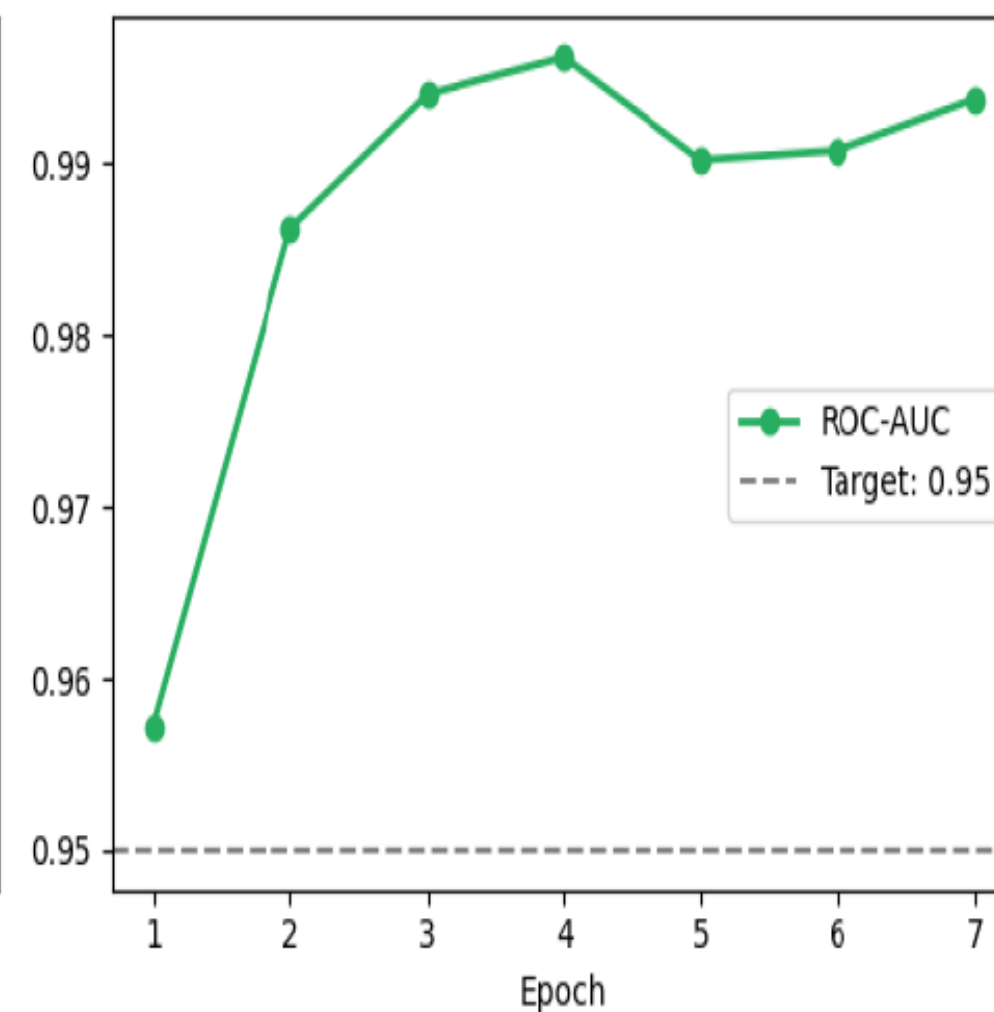
Training Loss



Validation F1 (Fraud Class)



Validation ROC-AUC

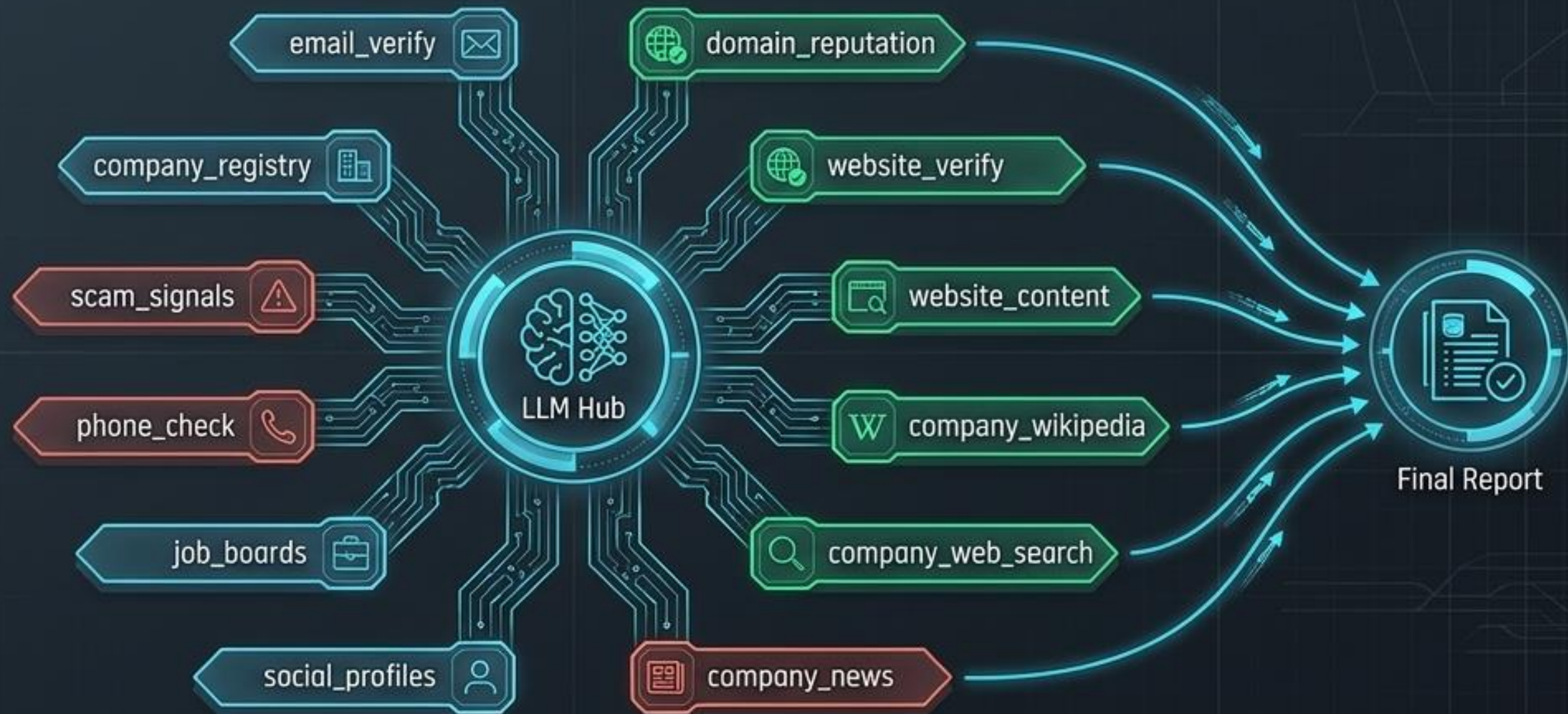


Diagnostic Insight

Automated Bayesian optimization (Optuna) successfully identified the exact learning rate and focal gamma values ($\gamma = 1.6920$) to stabilize the pipeline without manual guesswork, enabling stable convergence despite the severe 20:1 class imbalance.

Beyond Classification – 12-Tool Evidence Pipeline

Why a classifier isn't enough: It assigns probability, but it cannot verify external facts.



This produces human-readable evidence, not just a probability score.

Flask Web-App — Three Input Modes

Input Paths

1. **Paste text:**
Job description entered directly.
2. **File upload:**
PDF, DOCX, TXT, HTML, CSV.
3. **LinkedIn URL:**
Scrapes and analyzes live posting.

The screenshot shows a web application interface with a dark theme. On the left, there are three input sections: 'INPUT' (a large text area), 'File upload' (with an upload icon), and 'URL' (with a text field and a link icon). On the right, the results are displayed. At the top is a 'VERDICT: SUSPICIOUS (Confidence: 88%)' banner. Below it is a grid of 12 tool result cards, each with an icon and a brief description. At the bottom is a 'FULL LLM INVESTIGATION REPORT' section containing several paragraphs of generated text.

Tool	Result
EMAIL_VERIFY	Invalid Email, Score: 8.92
DOMAIN_REPUTATION	Blacklisted, 4/15 entries
WEBSITE_VERIFY	No HTTPS, Score: 6.55
SCAM_SIGNALS	Common keywords found, 5 hits
PHONE_CHECK	Voip detected, Location: NY
JOB_BOARDS	No active listings found
COMPANY_REGISTRY	Firm found in major databases
SOCIAL_PROFILES	Linked profiles: 2 profiles
WEBSITE_CONTENT	Content template found, 1 hit
COMPANY_WIKIPEDIA	No entry found
COMPANY_WEB_SEARCH	Found results, poor relevance
COMPANY_OENS	No public records found, 2...

What the User Sees

- Verdict banner (SAFE / **SUSPICIOUS** / LIKELY_FAKE) with confidence score.
- 12 tool result cards, each with 2–4 sentence LLM summary.
- Full LLM-written investigation report.

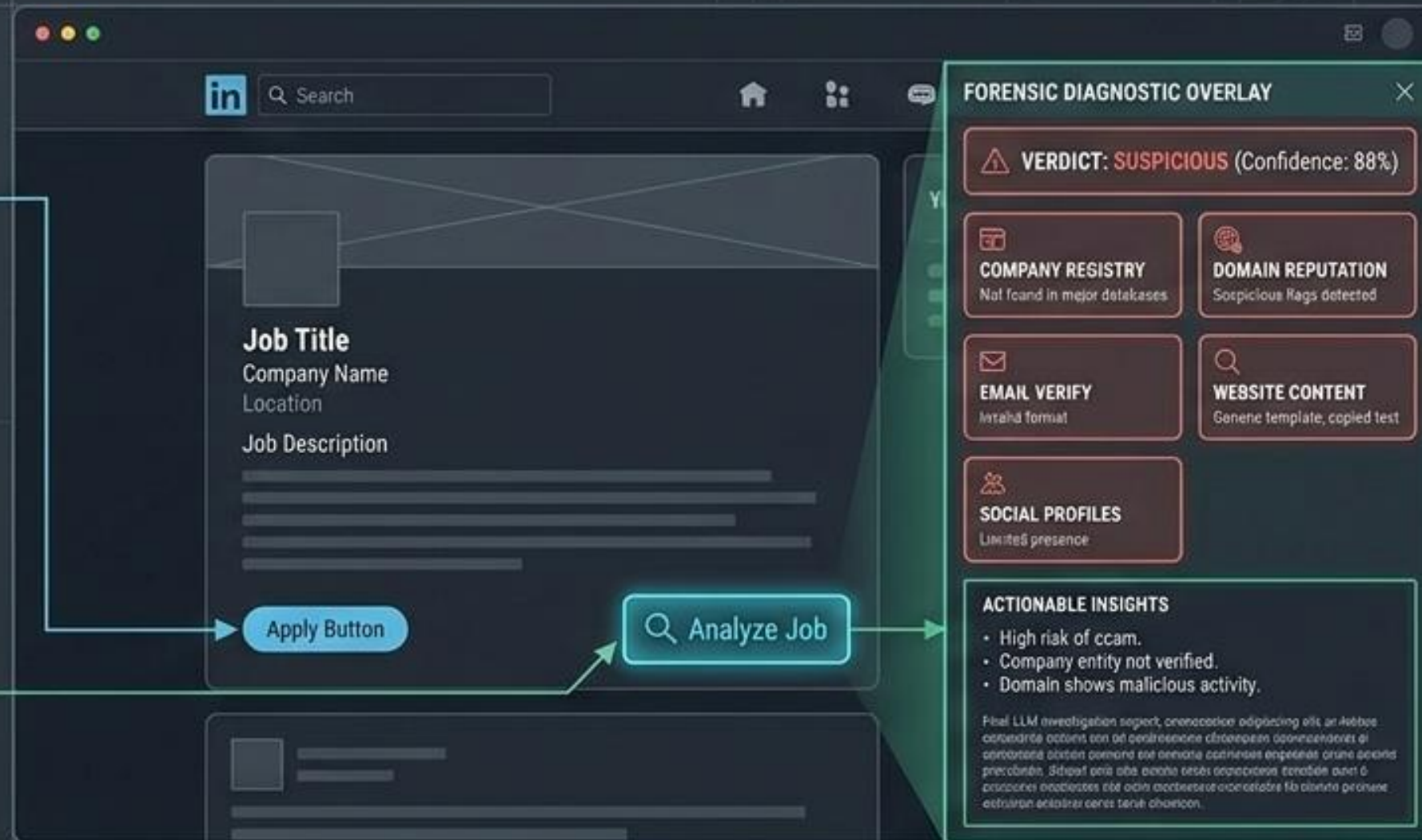
Real-Time Protection on LinkedIn

Bringing AI to the User

- Injects a floating “ **Analyze Job**” button directly into the LinkedIn DOM.
- Powered by Google Gemini API and Vanilla JS — zero server latency.

The Experience

- **1-Click Execution:** Scrapes active job details directly from the webpage.
- **Instant Verdict:** Generates a color-coded overlay panel within 3–5 seconds.
- **Actionable Insights:** Translates complex fraud signals into simple bullet points for the job seeker.



Team Contributions

Arun Dutta

- Led data preprocessing pipeline development.
- Synthesized literature review & gap analysis.
- Synthesized results for final reporting.

Hritik Roshan Maurya

- Engineered Agentic AI & LangChain orchestrator.
- Built 16-field extraction logic.
- Implemented Web-App REST API.

COLLABORATIVE
PROJECT
EXECUTION

Vivek Bajaj

- Directed ML pipeline strategy.
- Performed Optuna HPO (25 trials).
- Calibrated thresholds & deployed to HuggingFace.

Vishwas Mehta

- Developed Chrome Extension & Browser Integration.
- Implemented Metadata Anomaly Detector.
- Led LinkedIn DOM scraping efforts.